



MSC MECHANICAL ENGINEERING

Imaging in Fluid Dynamics

Experiment 1: Dye Visualization on Cylinder Wake and Hydrofoil

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1 Abstract

This experiment studies wake formation and flow separation using dye visualization in a water tunnel. Two cases were analyzed: the wake behind cylindrical bars of different diameters and the stall of a symmetric hydrofoil.

Four cylinders with diameters of 1 mm, 2 mm, 4 mm, and 8 mm were tested. Two flow velocities were set, corresponding to 10% and 15% of the tunnel capacity.

The mean fluid velocities were approximately 6.9605 cm/s and 9.0888 cm/s, respectively.

The Reynolds number ranged from 69.33 to 554.62 for the 10% case and from 90.53 to 724.21 for the 15% case.

The Strouhal number ranged from 0.125 to 0.19 and from 0.137 to 0.202 for the respective cases, showing the expected increase with Reynolds number.

The second part focused on a NACA 0015 hydrofoil. The angle of attack was gradually increased to observe the flow behavior. Flow separation and stall occurred between 5 ° and 10 °.

The graphical results showed a consistent correlation between the Reynolds and Strouhal numbers, in agreement with expected vortex shedding behavior. The experiment successfully visualized vortex shedding and boundary layer separation.

2 Introduction

2.1 Context and Motivation

The behavior of fluid flow around bodies is a key topic in fluid dynamics. When flow passes a cylinder, vortices are shed periodically, forming a Kármán vortex street. This process depends mainly on the Reynolds number, which compares inertial and viscous forces. The vortex-shedding frequency defines the Strouhal number, a key parameter for wake dynamics.

Flow around an airfoil shows a different behavior. At low angles of attack, the flow remains attached. Beyond a critical angle, the boundary layer separates, and stall occurs.

In this experiment, dye visualization was used in a water tunnel to study these two flows. Wakes behind four cylinders of different diameters were observed at two flow velocities to estimate shedding frequencies and Strouhal numbers. The stall angle of a symmetric hydrofoil was also determined from the onset of flow separation.

These measurements illustrate fundamental unsteady flow phenomena and their dependence on the Reynolds number and geometry.

2.2 Flow Physics and Wake Characterization

When a fluid flows around a solid body, such as a cylinder, the boundary layer separates from the surface and forms a region of recirculating flow downstream of the object. This region is known as the wake. At sufficiently high Reynolds numbers, the separated shear layers become unstable and alternate vortices are shed from each side of the body, forming a repeating pattern called the **von Kármán vortex street**, see Figure 1.



Figure 1: Visualization of the von Kármán vortex street behind a circular cylinder⁴

This behavior can be explained by differences in velocity and pressure around the cylinder. According to Bernoulli's principle, the total energy along a streamline remains constant and can be expressed as:

$$P + \frac{1}{2}\rho v^2 + \rho gh = \text{constant}$$

where P is the static pressure, ρ the fluid density, v the local velocity, and h the elevation. As the flow accelerates around the sides of the cylinder, its velocity increases, leading to a drop in static pressure. As we can see in the Figure 2, the resulting pressure imbalance between the front and rear of the cylinder causes flow separation and induces the alternating shedding of vortices. .

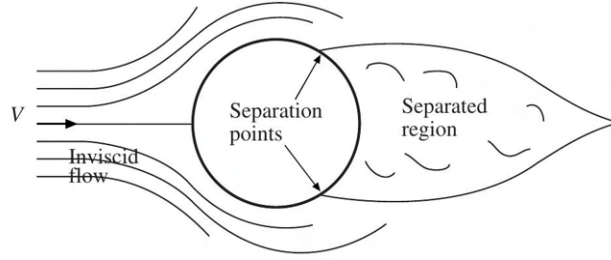


Figure 2: Flow separation behind a cylinder⁵

The behavior of the wake depends mainly on two dimensionless numbers. The **Reynolds number (Re)** is a dimensionless quantity used to predict flow patterns in different fluid flow situations. It expresses the ratio between inertial and viscous forces within a fluid and indicates whether the flow will be laminar, transitional, or turbulent. It is defined as:

$$Re = \frac{\rho v D}{\mu} = \frac{v D}{\nu} \quad \text{with} \quad \nu = \frac{\mu}{\rho}$$

where D is the diameter of the cylinder (m), v the characteristic velocity (m/s), ρ the fluid density (kg/m³), μ the dynamic viscosity (Pa·s), and ν the kinematic viscosity (m²/s). At low Re , viscous forces dominate, resulting in a smooth and steady flow. As Re increases (typically above $Re = 49$), inertial effects become significant, leading to periodic flow separation and vortex shedding.

⁴Wake from HandWiki: Physics.

⁵Flow separation from *Schaum's Outline of Fluid Mechanics* by Merle C. Potter and David C. Wiggert, McGraw-Hill Education, 2008.

The **Strouhal number (St)** is a dimensionless parameter that characterizes the periodicity of vortex shedding in unsteady flows. It represents the ratio between the time scale of vortex formation and the convective time scale of the flow, and is defined as:

$$St = \frac{fD}{v} = \frac{D}{\lambda} \quad \text{with} \quad f = \frac{v}{\lambda} = \frac{1}{T} = \frac{\Omega}{2\pi}$$

where f is the vortex-shedding frequency (Hz), Ω the angular frequency (rad/s) and λ the wavelength (m) of the vortex street. The Strouhal number evolves with the Reynolds number, capturing the transition from steady to unsteady and eventually turbulent wake regimes. This dependency provides a quantitative connection between flow stability and vortex shedding frequency, and serves as a key reference for comparing experimental and theoretical studies of unsteady wake flows.

2.3 Theoretical Considerations

As we have seen in class, the Strouhal number usually ranges between $St \approx 0.1 - 0.2$. It stays almost constant for a wide range of Reynolds numbers because the ratio between the vortex wavelength and the object diameter remains nearly the same. This means that the distance between vortices and their shedding frequency scale proportionally with the flow velocity. The example discussed in class showed that vortex shedding can occur over a wide range of Reynolds numbers, from about $200 < Re < 10^5$, with a Strouhal number remaining nearly constant around $St \approx 0.2$. This demonstrates the universality of the vortex shedding phenomenon, observed both at laboratory scale and in large-scale natural flows such as the wake behind Guadalupe Island in the Pacific Ocean.

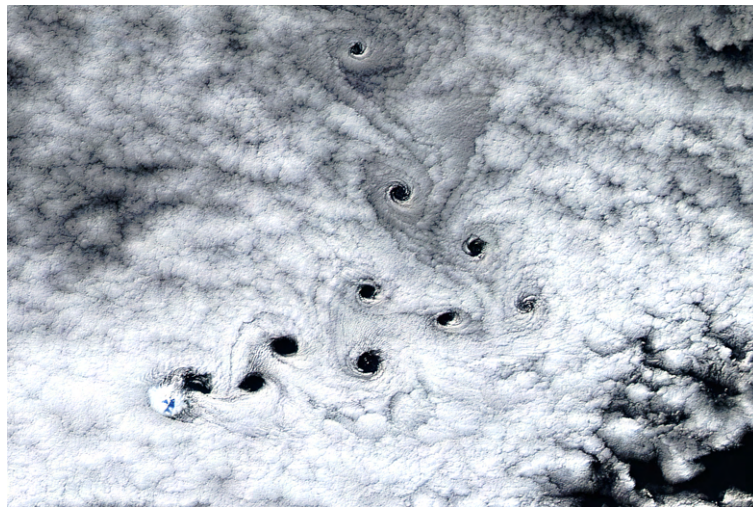


Figure 3: Kármán vortex street on Heard Island⁶

1. **Laminar steady regime** ($Re < 49$)
2. **Laminar vortex shedding** ($49 < Re < 190$)
3. **3D wake transition to turbulence** ($190 < Re < 260$)
4. **Disordered wake with fine-scale structures** ($260 < Re < 1000$)

⁶Image by Jeff Schmaltz from *Nasa Earth Observatory*

When the Reynolds number is low, the flow is laminar and separation is almost nonexistent; it is so small that it does not significantly affect the flow behind the body. For higher Reynolds numbers, in the transitional or turbulent regime, the flow clearly separates from the surface and a distinct wake forms behind the body. This separation creates a low-pressure region near the rear of the object. According to Bernoulli's principle discussed in Section 2.2, an increase in pressure along the streamline corresponds to a decrease in velocity. In some cases, this pressure difference is large enough to reduce the velocity to zero or even produce reversed flow. As a result, the fluid behind the object may appear almost stationary. This stagnant region, characterized by negative pressure, is typically located directly behind the blunt body, as shown in Figure 4.

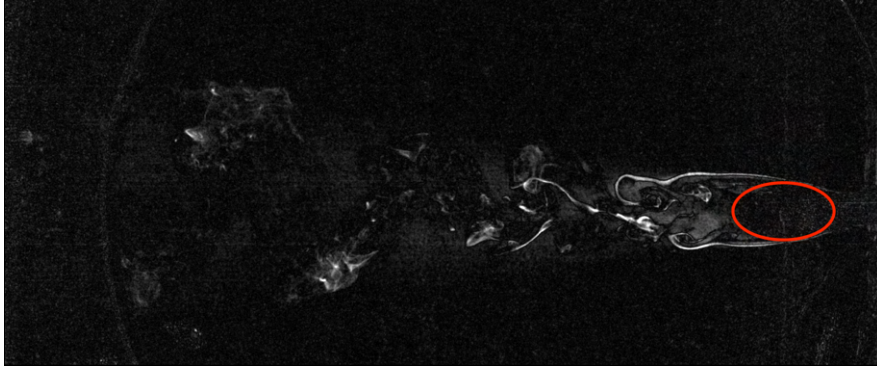


Figure 4: Steady region behind an 8mm cylinder

2.4 Aims of the Study

We will analyze the wake geometry such as the wavelength λ and the frequency of oscillation f . The spatial wavelength of the vortex street represents the distance separating two consecutive vortices on the same side of the wake, clearly visible in the experimental figure 5. Together, the wavelength and shedding frequency provide a complete description of the unsteady behavior of the wake and allow the determination of the Strouhal number, which links the temporal and spatial characteristics of vortex shedding.

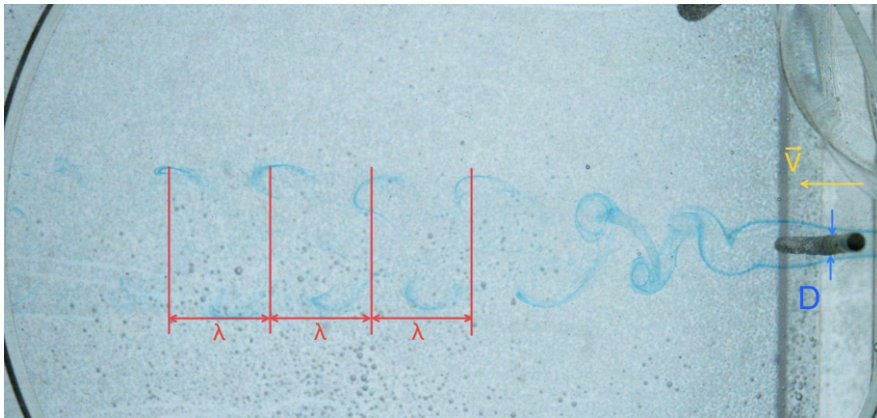


Figure 5: Periodic vortex formation in the third cylinder's wake at low velocity

The first objective of this study is to quantify these wake characteristics by extracting f and λ from the image analysis. This will be done using MATLAB for filtering, scaling, and velocity estimation,

allowing the calculation and comparison of Reynolds and Strouhal numbers into a plot.

A second objective is to examine the flow around a NACA 0015 hydrofoil by varying its angle of attack (AoA) and identifying the stall point corresponding to flow separation.

3 Experimental Setup

The experiment was carried out in a recirculating water tunnel designed for flow visualization, as shown in Figure 12. A honeycomb section was placed at the tunnel inlet to straighten the incoming flow and reduce turbulence. A dye injector was installed along the center line of the tunnel to visualize the wake structures generated by the different objects studied.

Four circular cylinders with diameters of 1, 2, 4, and 8 mm were tested, together with a symmetric NACA 0015 hydrofoil. Both configurations were mounted on transparent rotating disks positioned in the center of the test section. This setup allowed for quick interchange between the cylinders and for manual adjustment of the angle of attack of the hydrofoil. The angle was progressively increased until the onset of stall.

A CMOS (complementary metal-oxide-semiconductor) camera of 12 megapixels resolution was placed above the water tunnel, aligned with the location of the test body. The Figure 8 show this top view that provides a clear visualization of the dye perturbations and vortex formation in the wake. The recordings consisted of short sequences of 3 to 10 seconds at 100 fps with an exposure time of 200 μ s and a size of 1920 x 1080 pixels. The tunnel was adjusted to the selected flow condition (10 % or 15 %) and run until steady-state conditions were reached, typically after 10 to 15 minutes.



Figure 6: Water tunnel setup with honeycomb and dye injector.

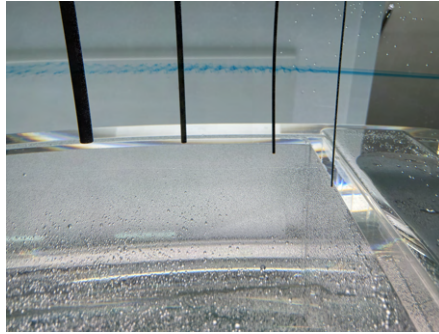


Figure 7: Test cylinders in the test section.

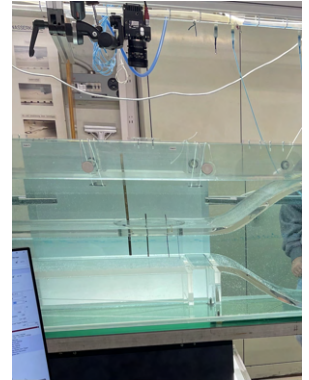


Figure 8: Camera view-point above the test section.

4 Results and Data Analysis

4.1 Image Calibration and Velocity Estimation

To calibrate the images, a chessboard target was used with squares of 1 cm by 1 cm, as shown in Figure 13. Using the MATLAB *Image Calibration* script, the number of pixels corresponding to each edge of a square was identified. This provided a spatial conversion ratio of 0.01264 cm/px.

The calibration accuracy depends on the corner detection of the chessboard. We can estimate a detection error of about ± 0.5 pixel per edge, which lead to an uncertainty of approximately ± 0.00015 cm/px. For the first three cylinders, a plexiglass block was placed on top of the test section to reduce surface ripples. The difference in calibration with and without the block was only 0.00001 cm/px, confirming that its effect is negligible.

To estimate the flow velocity, a MATLAB tracking script was used to follow a dye perturbation by selecting a region of interest (ROI). The Figure 9 presents the script that tracks the displacement of a ROI in both x and y directions across all frames, providing a total displacement $(\delta x, \delta y)$ over a number of frames N . With this method, we also obtain the angle of displacement of the velocity fluid. The flow velocity v is then computed using the following expression:

$$v = \frac{f_s \times D \times r}{N} \quad [cm/s]$$

where:

- N is the number of frames,
- f_s is the frame rate (100 fps),
- D is the total displacement in pixels,
- r is the spatial ratio (0.01264 cm/px).

The theoretical and measured flow velocities for the two operating conditions are summarized below.

Table 1: Comparison between theoretical and measured flow velocities.

	10% case	15% case
Number of frames N	351	266
Frame rate f_s (fps)	100	100
Spatial ratio r (cm/px)	0.01264 ± 0.00015	0.01264 ± 0.00015
Displacement D (px)	1920	1920
Theoretical velocity v_{th} (cm/s)	6.9 ± 0.1	9.1 ± 0.1
Measured velocity v_{code} (cm/s)	6.9605	9.0888

This approach provides an accurate estimation of the mean flow velocity within the field of view, based on the motion of visible flow perturbations, but several sources of uncertainty remain. The flow velocity is not perfectly constant during the entire experiment, but it is measured only once at the beginning of the protocol and assumed constant for all recordings. This can introduce small variations between measurements.



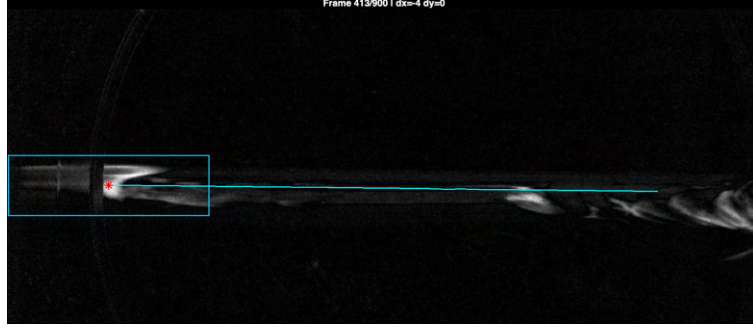


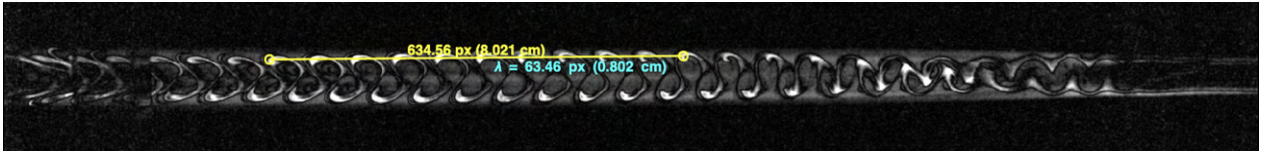
Figure 9: Tracking of a dye perturbation

4.2 Frequency and Wavelength Determination

The vortex shedding frequency was obtained from the temporal variation of light intensity at a chosen point in the wake. The mean intensity of a small region around this point was analyzed through a Fourier transform to identify the dominant frequency f_{dom} , corresponding to the periodic release of vortices in the Kármán vortex street.

The spatial wavelength λ of the vortex street was determined in two different ways. First, the theoretical wavelength was calculated using the relation $\lambda = U/f_{\text{dom}}$, where U is the mean flow velocity and f_{dom} the dominant shedding frequency.

Second, the wavelength was measured directly from the flow images. Two points were selected along the same side of the wake, corresponding to a known number of complete vortex periods that we can directly count on the Figures 23 and 24. The total distance between the two points was divided by the number of periods to obtain the wavelength, then converted from pixels to centimeters using the calibration ratio of 0.01264 cm/px.

Figure 10: Measurement of the vortex street wavelength λ of the small cylinder at 10%.

4.3 Hydrofoil Stall Angle Analysis

The angle of attack (AoA) of the NACA 0015 hydrofoil was measured directly from the recorded images using an interactive MATLAB tool. Two points were selected along the chord line of the hydrofoil and the corresponding angle relative to the horizontal was calculated. The flow angle was determined separately using the velocity tracking script, which provided the mean direction of the dye displacement. This angle was found to be very small, about 0.6° , and was therefore considered negligible in the AoA calculation.

The procedure was repeated for 4 hydrofoil orientations to observe how the flow structure evolves with increasing AoA (Figures 25). At small angles, the flow remained attached to both surfaces of the foil, producing a symmetric wake. As the AoA increased, flow separation appeared on the

suction side and the wake became asymmetric with visible recirculation behind the trailing edge.

The first flow separation appears approximately at 6° . The stall angle was estimated to occur around 10° , where a clear and stable separation of the flow was observed and the periodic vortex pattern in the wake disappeared.

4.4 Summary of Quantitative Results

Based on the measured mean flow velocities and the cylinder diameters, the Reynolds numbers were calculated according to the definition given in Section 2.2. The mean velocities of 6.9605 cm/s and 9.0888 cm/s, corresponding respectively to 10% and 15% of the tunnel capacity, were used for the calculations. The results for the four cylinders are shown below.

Table 2: Calculated Reynolds numbers for each cylinder diameter at 10% and 15% of tunnel capacity.

Cylinder Diameter D (mm)	$Re_{10\%}$	$Re_{15\%}$
1	69.33	90.53
2	138.66	181.05
4	277.31	362.10
8	554.62	724.21

As expected, the Reynolds number increases linearly with both the cylinder diameter and the flow velocity. The values range from approximately 70 to 730, covering laminar, transitional, and disordered flow regimes. According to the classification discussed in the lecture, vortex shedding remains laminar and periodic for $49 < Re < 190$, transitions to turbulence for $190 < Re < 260$, and becomes increasingly disordered for $Re > 260$. Once the flow regimes were identified through the Reynolds numbers, the corresponding vortex shedding frequencies and wavelengths were determined to further characterize the wake dynamics.

Table 3: Measured vortex shedding frequencies for each cylinder diameter at 10% and 15% of tunnel capacity.

Diameter D (mm)	Frequency at 10% (Hz)	Frequency at 15% (Hz)
1	8.753	12.465
2	5.317	7.872
4	3.000	3.738
8	1.654	2.302

Table 4: Comparison between theoretical and measured wavelength λ at 10% and 15%.

D (mm)	$\lambda_{10\%,th}$ (cm)	$\lambda_{10\%,meas}$ (cm)	$\lambda_{15\%,th}$ (cm)	$\lambda_{15\%,meas}$ (cm)
1	0.795	0.8021	0.729	0.7220
2	1.309	1.3363	1.154	1.1734
4	2.320	2.4250	2.431	2.0569
8	4.208	4.5487	3.948	4.0310

The error margin for the wavelength at the 8 mm cylinder is more important because the vortex shedding was very irregular at this scale, corresponding to the turbulent wake regime. Once the vortex shedding frequency and the corresponding wavelength were obtained, the Strouhal number St could be calculated using both definitions in Section 2.2.

These two formulations are theoretically equivalent, but experimental uncertainties in measuring the wavelength or the shedding frequency can lead to small discrepancies between the two.

Table 5: Comparison of Strouhal numbers calculated from measured frequency and wavelength at 10% and 15% of tunnel capacity.

D (mm)	$St_{10\%,f}$	$St_{10\%,\lambda}$	$St_{15\%,f}$	$St_{15\%,\lambda}$
1	0.126	0.125	0.137	0.137
2	0.153	0.149	0.173	0.171
4	0.172	0.165	0.165	0.194
8	0.190	0.176	0.202	0.196

The two calculation methods give consistent results except for the 8mm diameter cylinder at 10% of the tunnel capacity, confirming that both the frequency-based and the wavelength-based analyses accurately describe the vortex shedding dynamics within the expected Strouhal range of 0.1–0.2.

5 Discussion

5.1 Assumption and source of error

We made some assumptions in our experiment that are not negligible. We assumed the calculation of the Reynold number with a constant kinematic viscosity $\nu = 1.004 \times 10^{-6} \text{ m}^2/\text{s}$ of the water at 20°C , but we did not take into account the dye parameters. The pump can generate small mechanical vibrations that can slightly disturb the flow and the dye trajectory, which can cause a small noise that we try to avoid when we do the FFT.

In addition, about 500 frames were recorded at a rate of 100 fps, which ensures good temporal resolution. Therefore, image acquisition itself does not represent a significant source of uncertainty. Considering that the precision of the caliper is 0.1 mm, we can have an actual uncertainty on the diameter that is approximately $\pm 0.05 \text{ mm}$. The measurement of the wavelength has a large error margin because the points used for the measurement are selected manually, which introduces a potential positioning error (human error). This uncertainty is less significant for smaller cylinders, where a larger number of vortex periods can be sampled, providing a more reliable estimate of the mean wavelength as we do in Section 8.

5.2 Wake Structure and Strouhal–Reynolds Correlation

The dye visualizations showed a clear evolution of the wake with increasing Reynolds number. For small cylinders ($D = 1\text{--}2 \text{ mm}$, $Re < 200$), the flow remained laminar and periodic, forming a well-defined von Kármán vortex street. As Re increased ($200 < Re < 500$), the wake widened and lost coherence, and for the largest cylinder ($D = 8 \text{ mm}$, $Re > 500$), the flow became turbulent with irregular, rapidly merging vortices, consistent with the three-dimensional instabilities reported by Williamson⁷.

⁷C. H. K. Williamson, *Vortex dynamics in the cylinder wake*, *Annual Review of Fluid Mechanics*, Vol. 28, 1996.

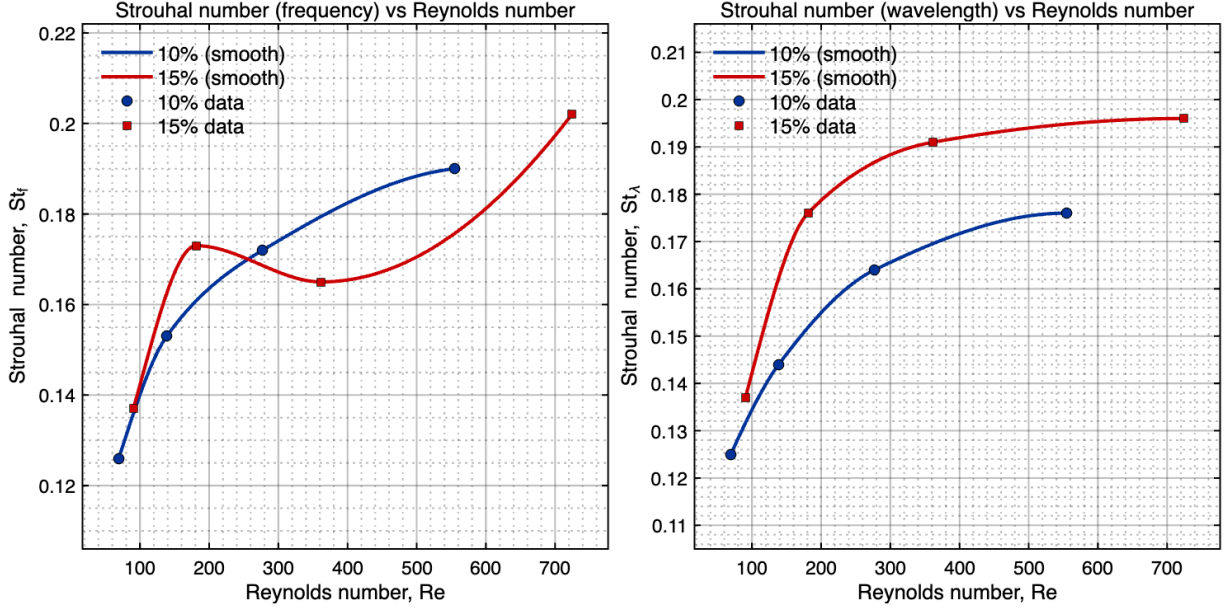


Figure 11: Strouhal–Reynolds number correlation at 10% and 15% flow conditions.

Almost all curves show a logarithmic increase with Reynolds number, consistent with the expected behavior reported in the literature. The Strouhal values, ranging from 0.1 to 0.2, match those typically observed for circular cylinders in laminar and early transitional regimes. In the frequency-based analysis, the 15% curve shows a slight drop at the third point, probably due to minor variations in mean flow velocity. The Strouhal numbers obtained from wavelength analysis are slightly lower than those from frequency, suggesting small overestimations in wavelength or underestimations in frequency, as discussed in Section 5.1.

6 Conclusions

This experiment showed the evolution of wake formation and flow behavior around circular cylinders and a NACA 0015 hydrofoil using dye visualization in a water tunnel. The relationship between the Strouhal and Reynolds numbers followed the expected theoretical trend, with St tending toward a limit near 0.2 as Re increased. The wake evolved from laminar and periodic to irregular and turbulent, consistent with theoretical predictions.

For the hydrofoil, the flow remained attached up to an angle of attack of approximately 10° , beyond which stall occurred, in agreement with the expected behavior of a symmetric airfoil.

Several limitations were identified in the experimental approach. The mean flow velocity was assumed constant throughout all recordings, which may have introduced uncertainty. Measuring it before each acquisition or enabling simultaneous tracking could improve accuracy. The wavelength estimation was also affected by manual point selection, especially at higher Reynolds numbers where vortices became less coherent. Enhancing image contrast and improving the measurement procedure would reduce this uncertainty.

Overall, we highlighted the main characteristics of wake dynamics and their dependence on the Reynolds number, while also revealing areas where measurement precision could be improved. Future studies could benefit from non-intrusive methods such as Particle Image Velocimetry (PIV), which allows particle tracking without direct intervention during the experiment and provides more accurate flow field measurements.

7 References

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8 Appendix

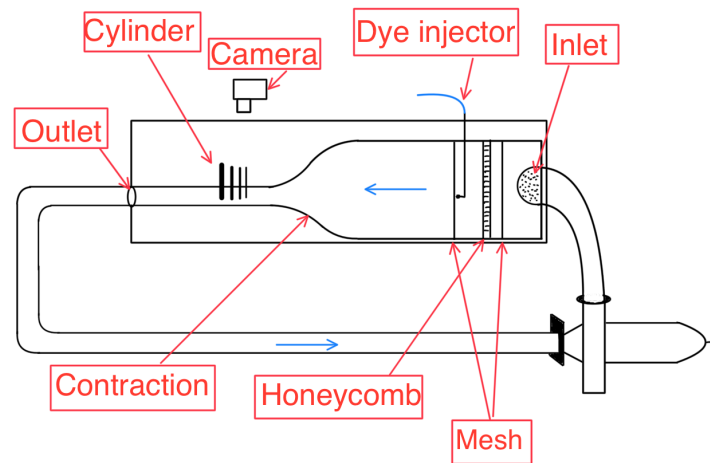


Figure 12: Sketch of the water tunnel

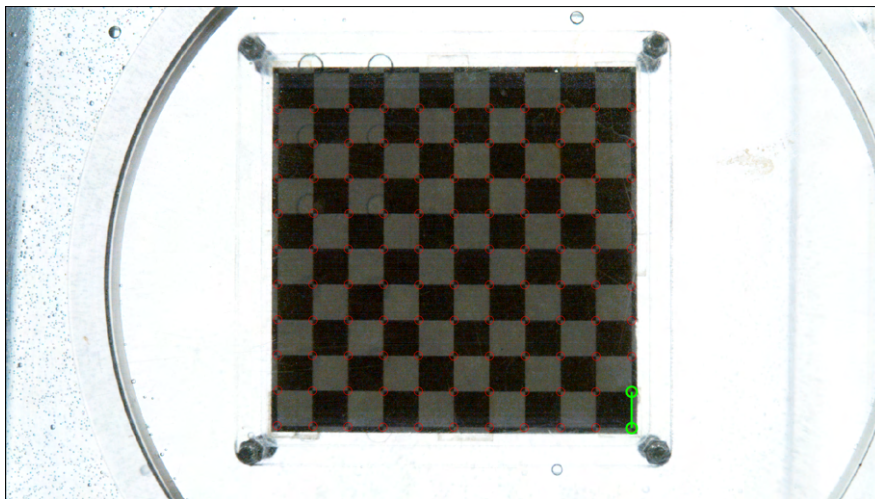


Figure 13: Calibration pattern using a 1 cm x 1 cm chessboard

Here we will enumerate the power frequency and intensity graphs for each case to show the principal frequencies given by the Matlab script we used. To improve the analysis of the 8 mm cylinder wake, an additional contrast filter was applied on top of the existing image processing. This secondary filtering enhanced the visibility of the dye structures by isolating only the regions of maximum intensity.

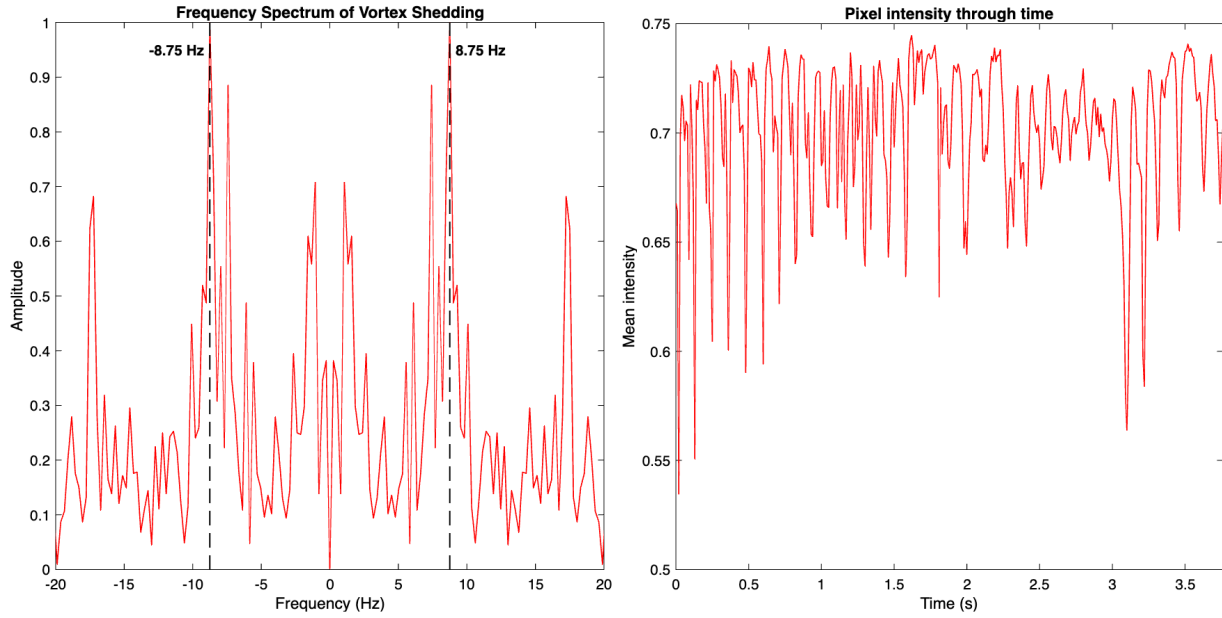


Figure 14: Vortex shedding frequency and intensity for the 1 mm cylinder at low velocity

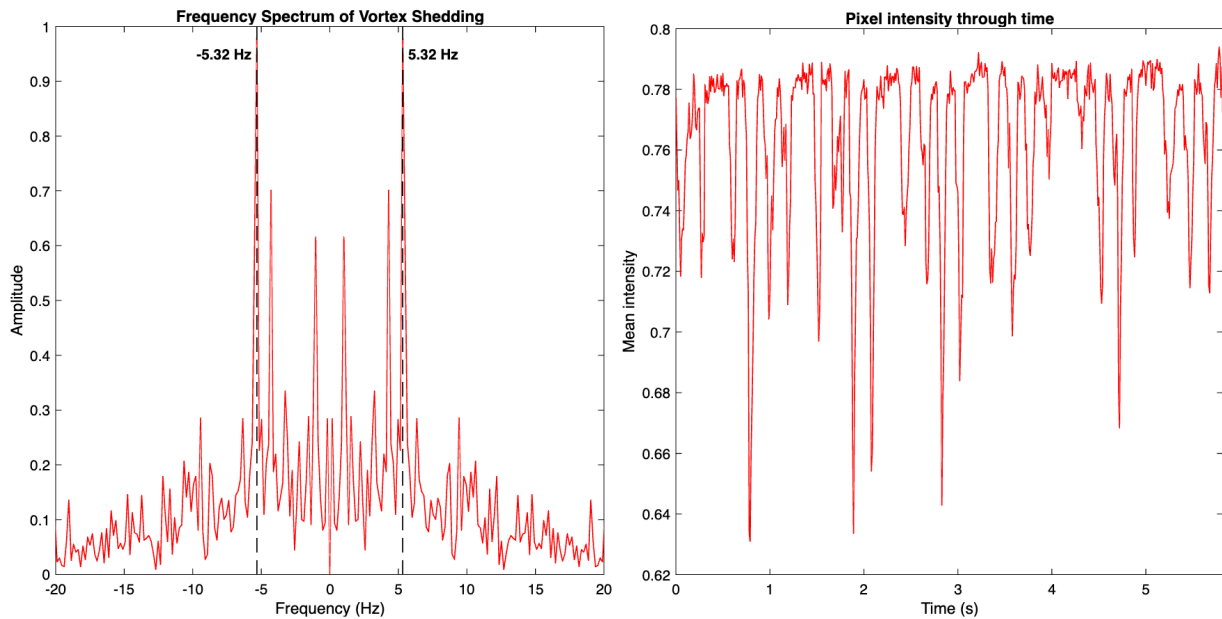


Figure 15: Vortex shedding frequency and intensity for the 2 mm cylinder at low velocity

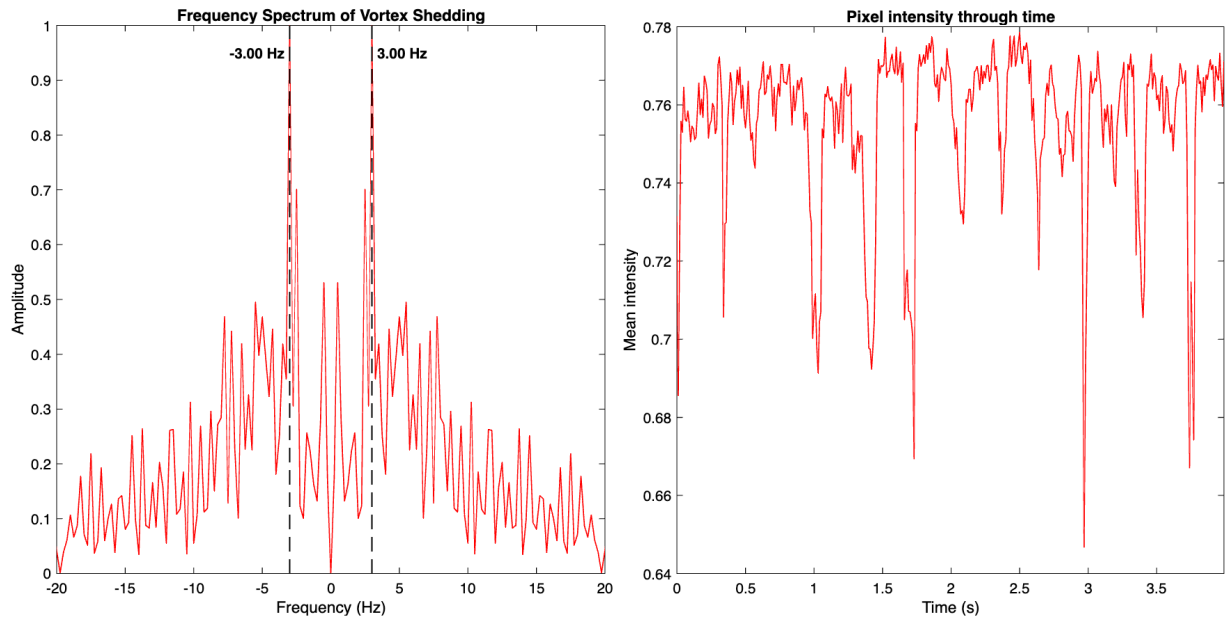


Figure 16: Vortex shedding frequency and intensity for the 4 mm cylinder at low velocity

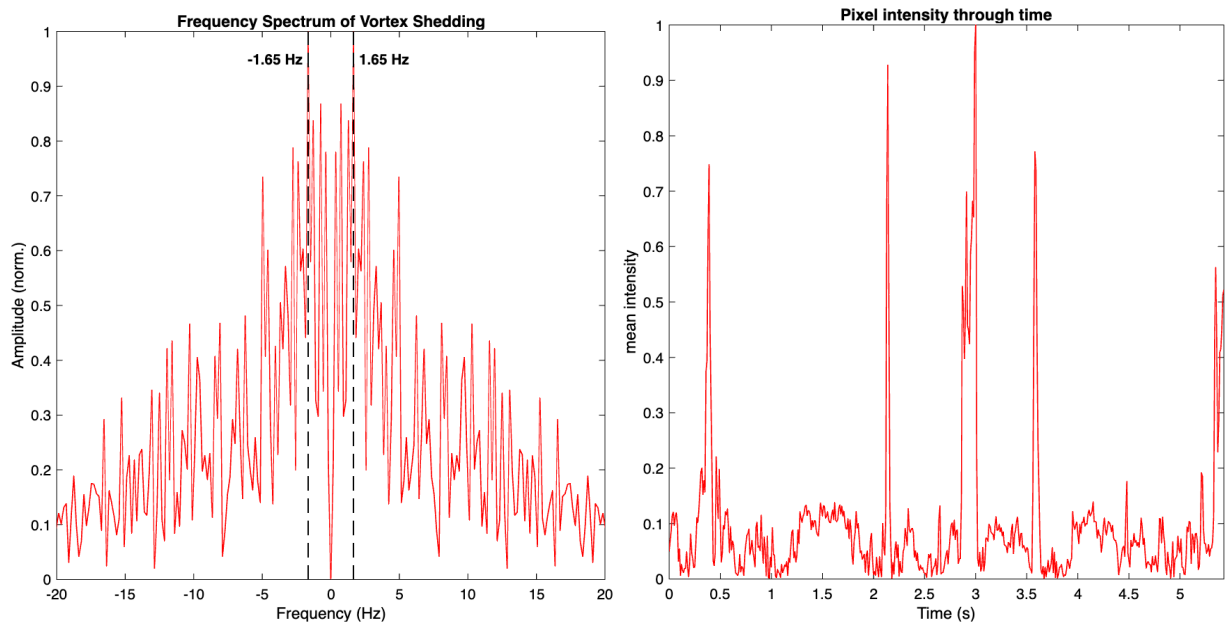


Figure 17: Vortex shedding frequency and intensity for the 8 mm cylinder at low velocity

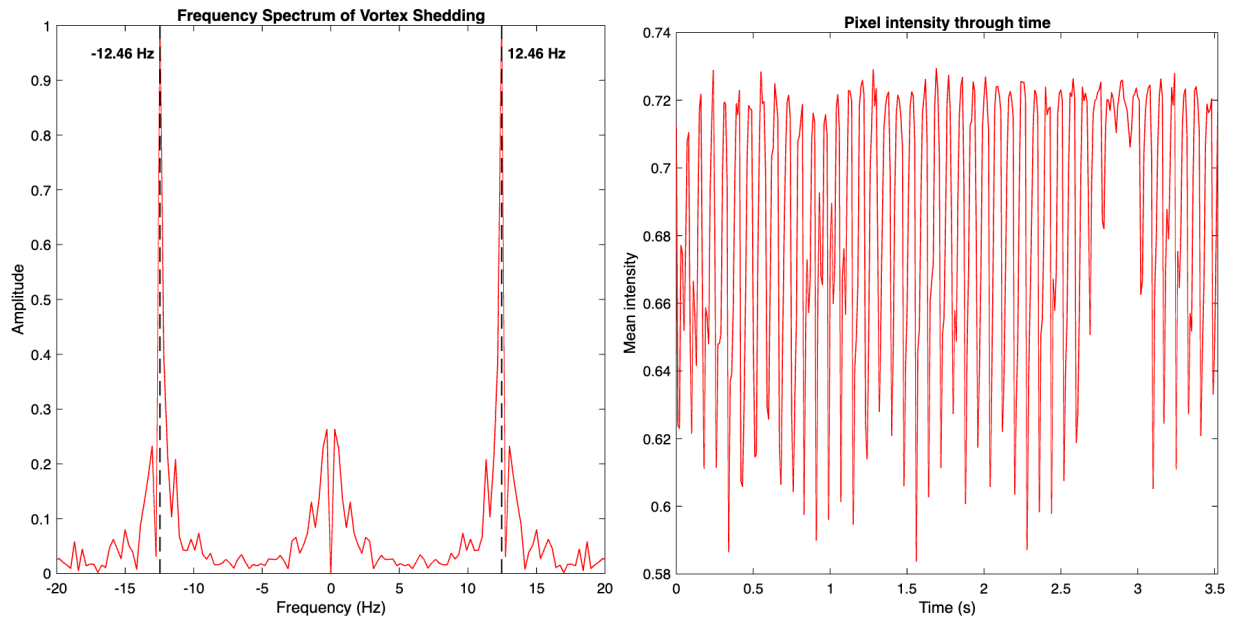


Figure 18: Vortex shedding frequency and intensity for the 1 mm cylinder at high velocity

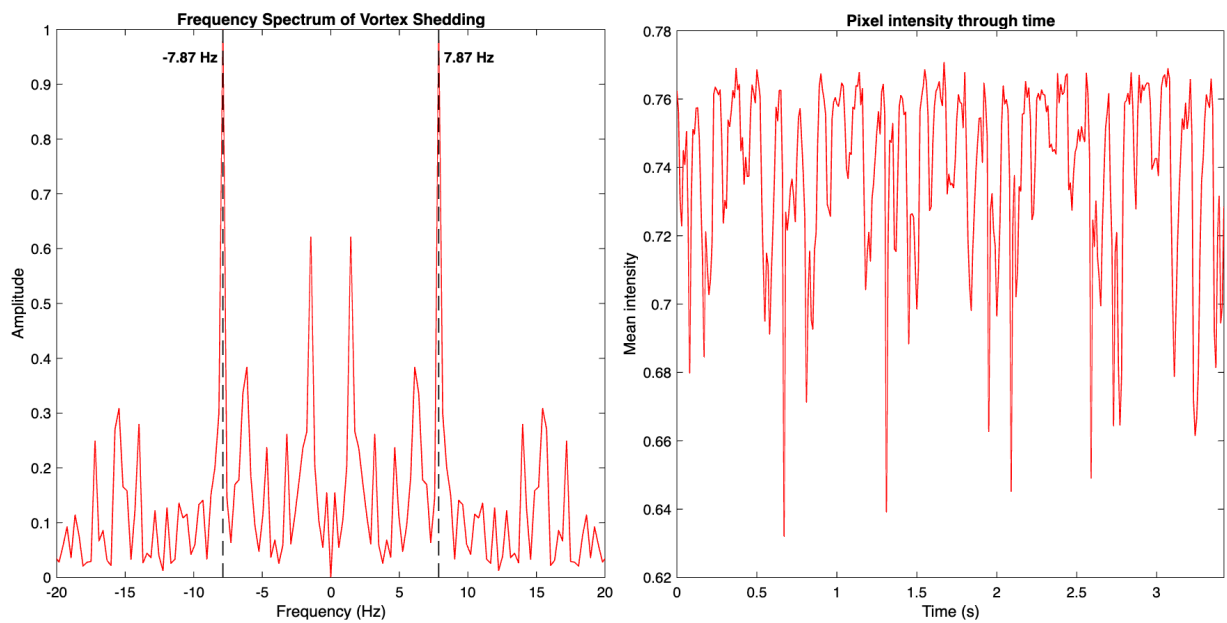


Figure 19: Vortex shedding frequency and intensity for the 2 mm cylinder at high velocity

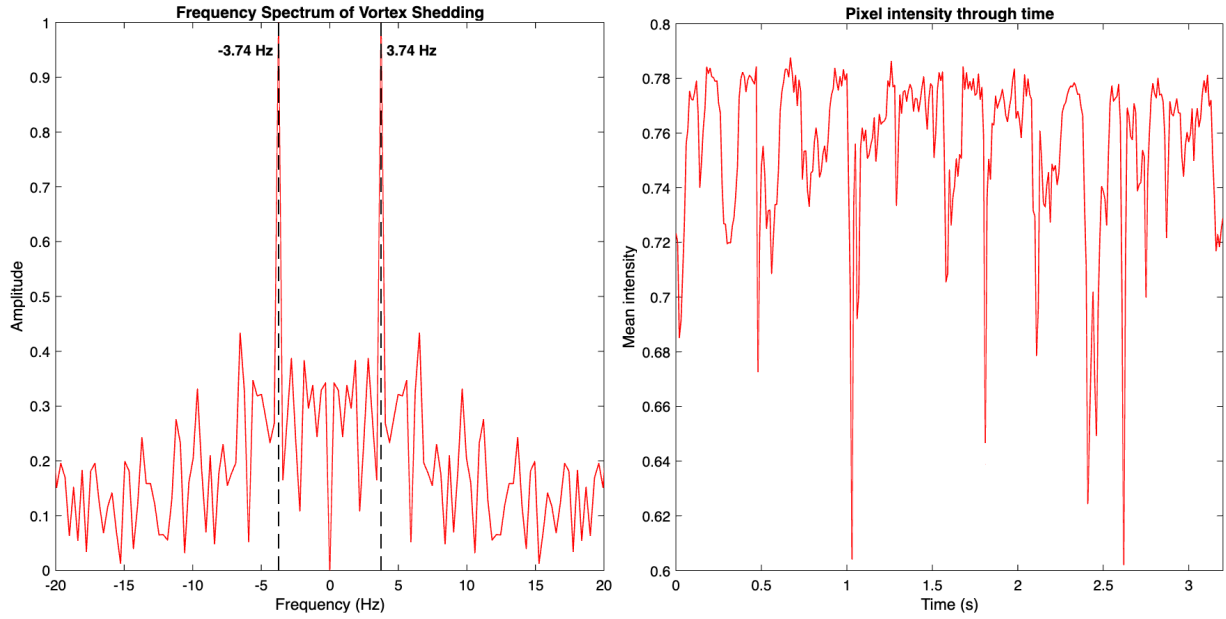


Figure 20: Vortex shedding frequency and intensity for the 4 mm cylinder at high velocity

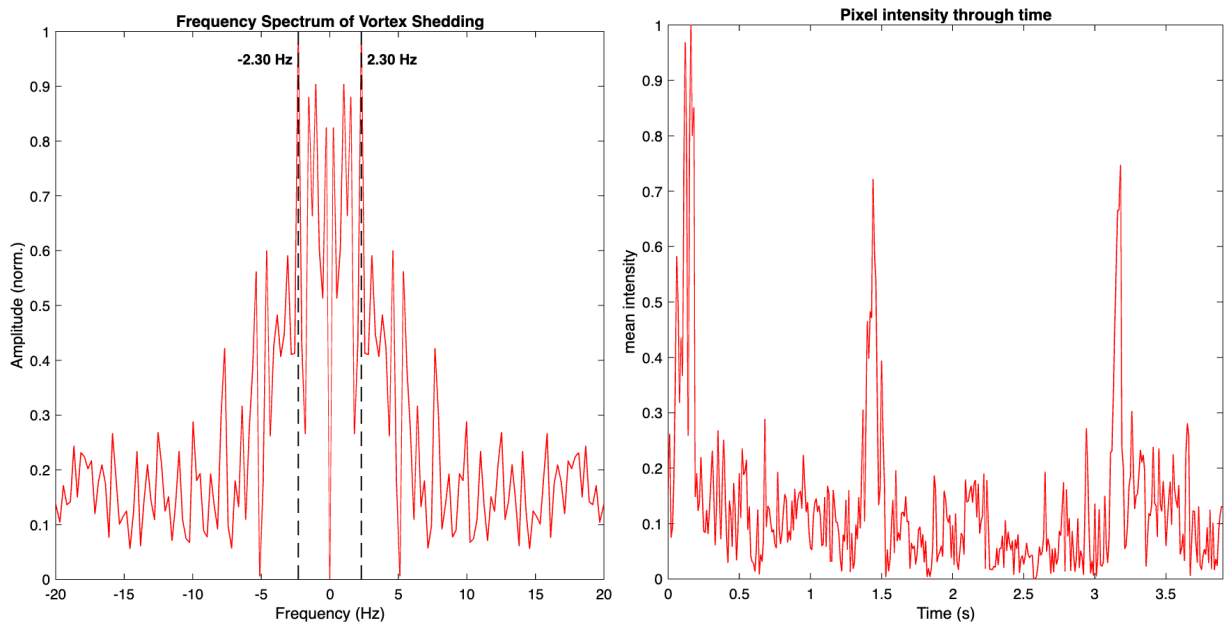


Figure 21: Vortex shedding frequency and intensity for the 8 mm cylinder at high velocity

The following figures present the wavelength measurements performed on the different cylinders. These visualizations illustrate how the periodic spacing of the vortices was used to determine the wavelength. The periodic pattern of the von Kármán vortex street can still be distinguished, allowing a rough estimation of the wavelength for the 8 mm cylinder.

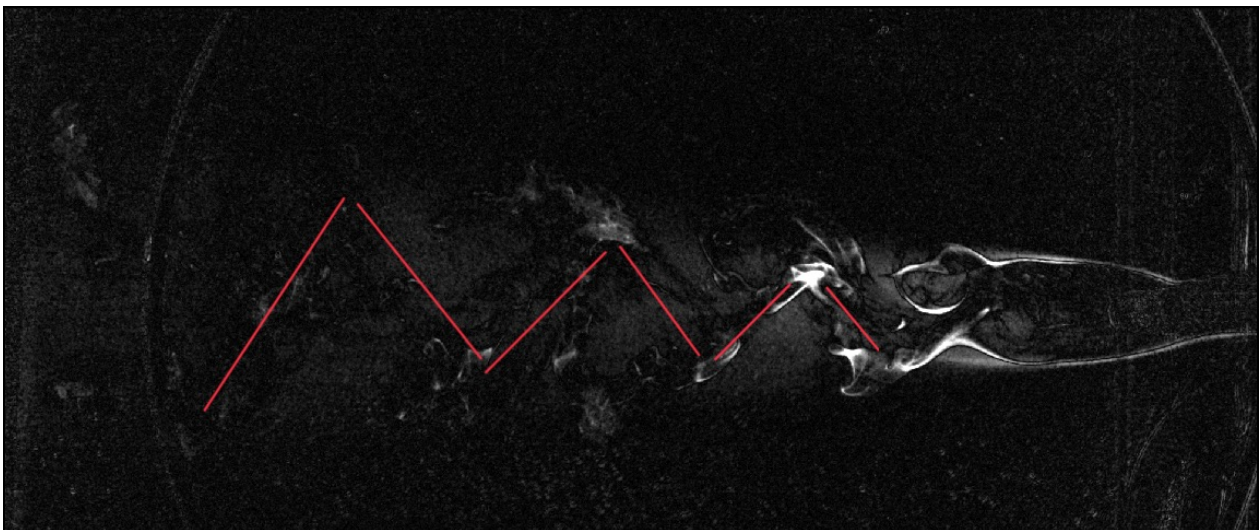
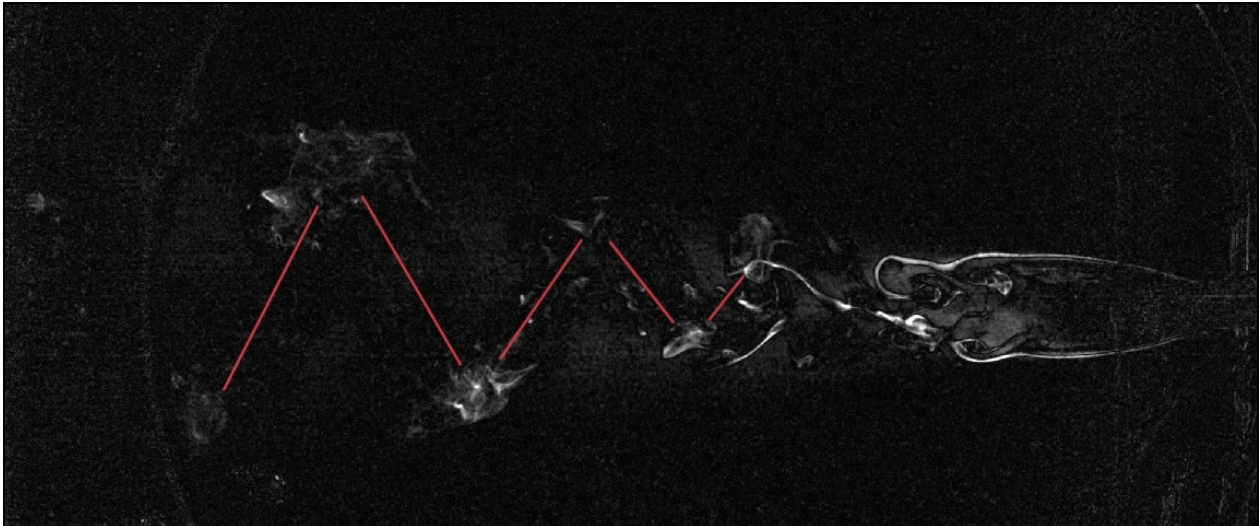


Figure 22: Approximation of the Kármán vortex street behind the 8mm cylinder

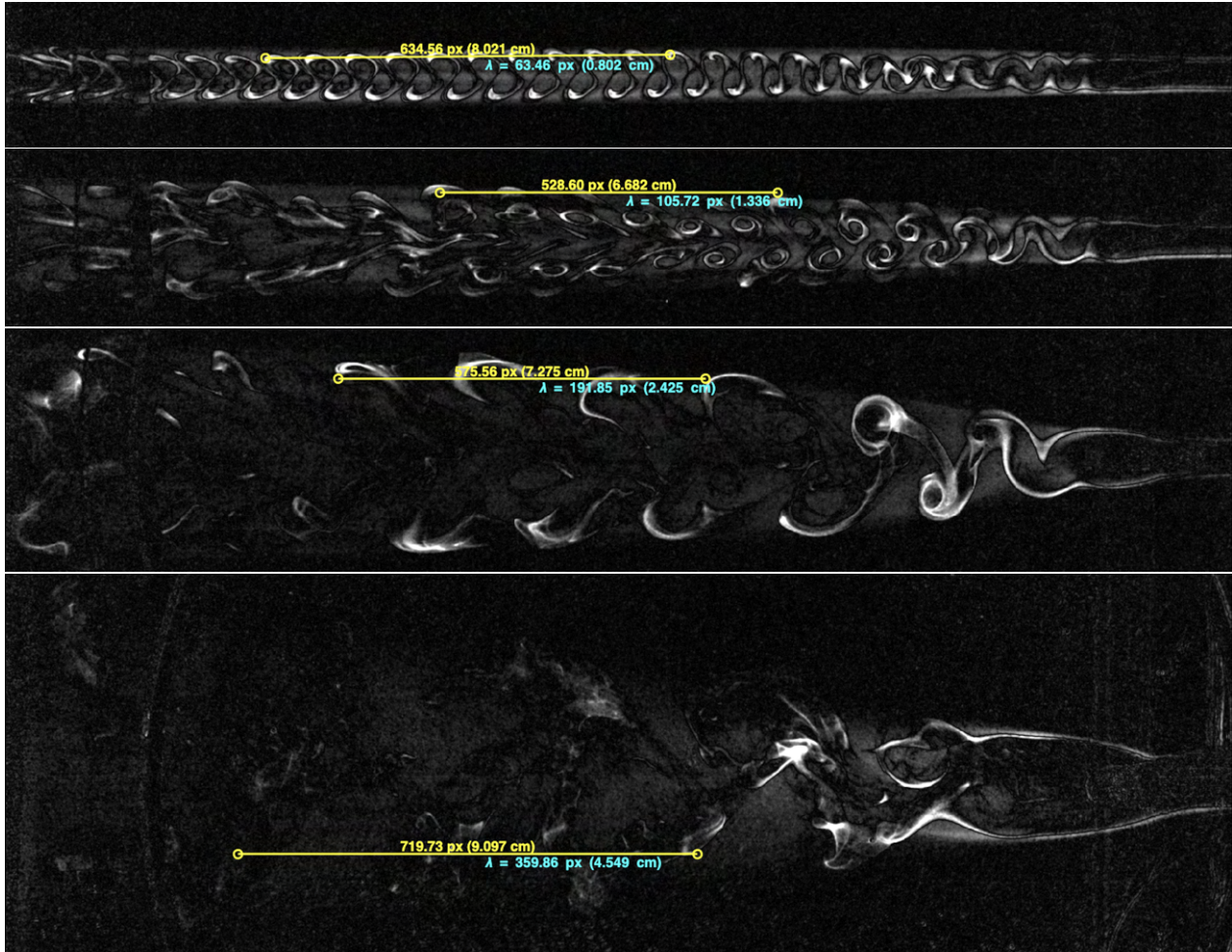


Figure 23: Wavelength measurement at $v_{10\%} \approx 6.9 \text{ cm/s}$

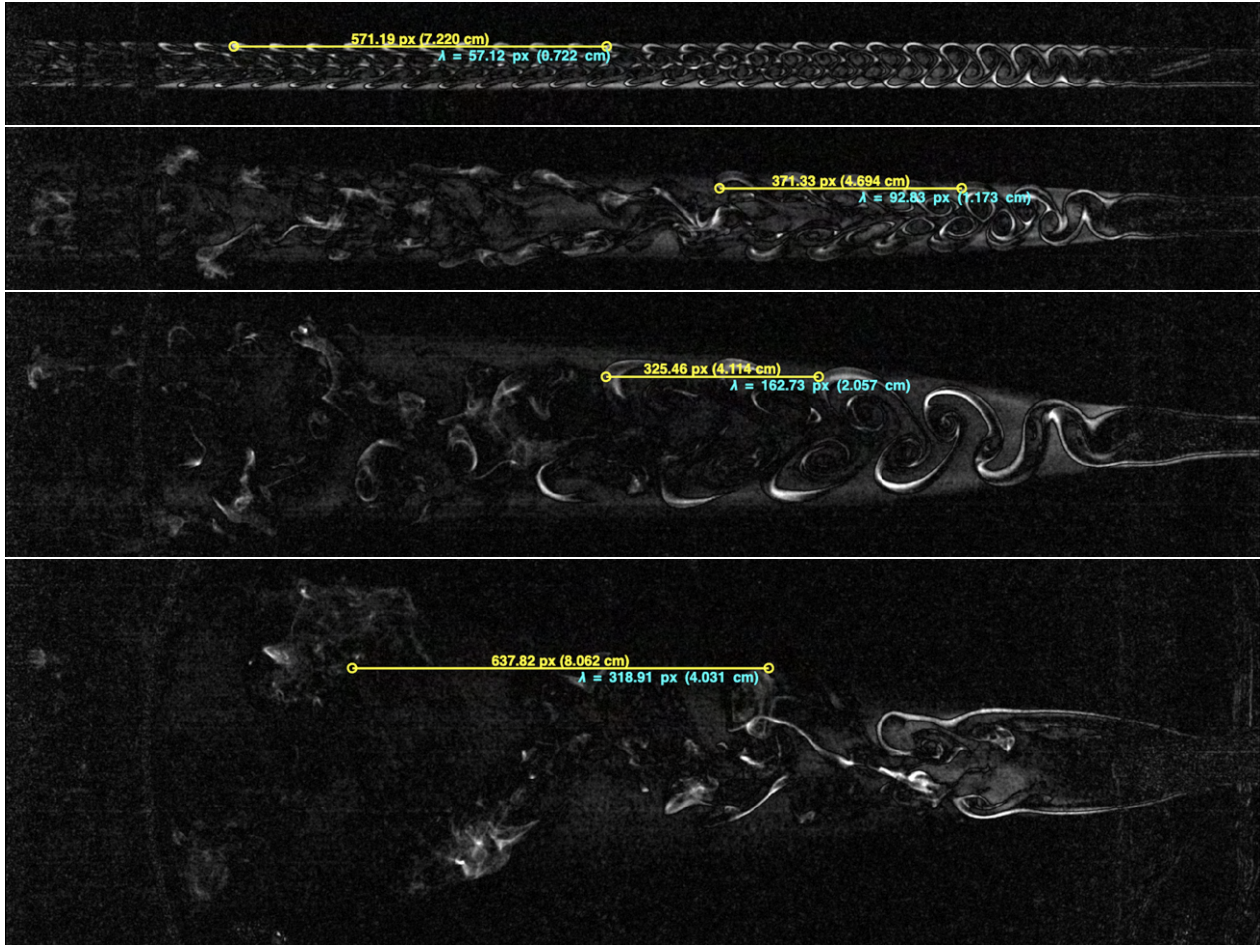


Figure 24: Wavelength measurement at $v_{15\%} \approx 9.1 \text{ cm/s}$

Finally, the following figures present the successive increments of the angle of attack for the NACA 0015 hydrofoil. These visualizations allow the identification of the flow separation and the stall angle.

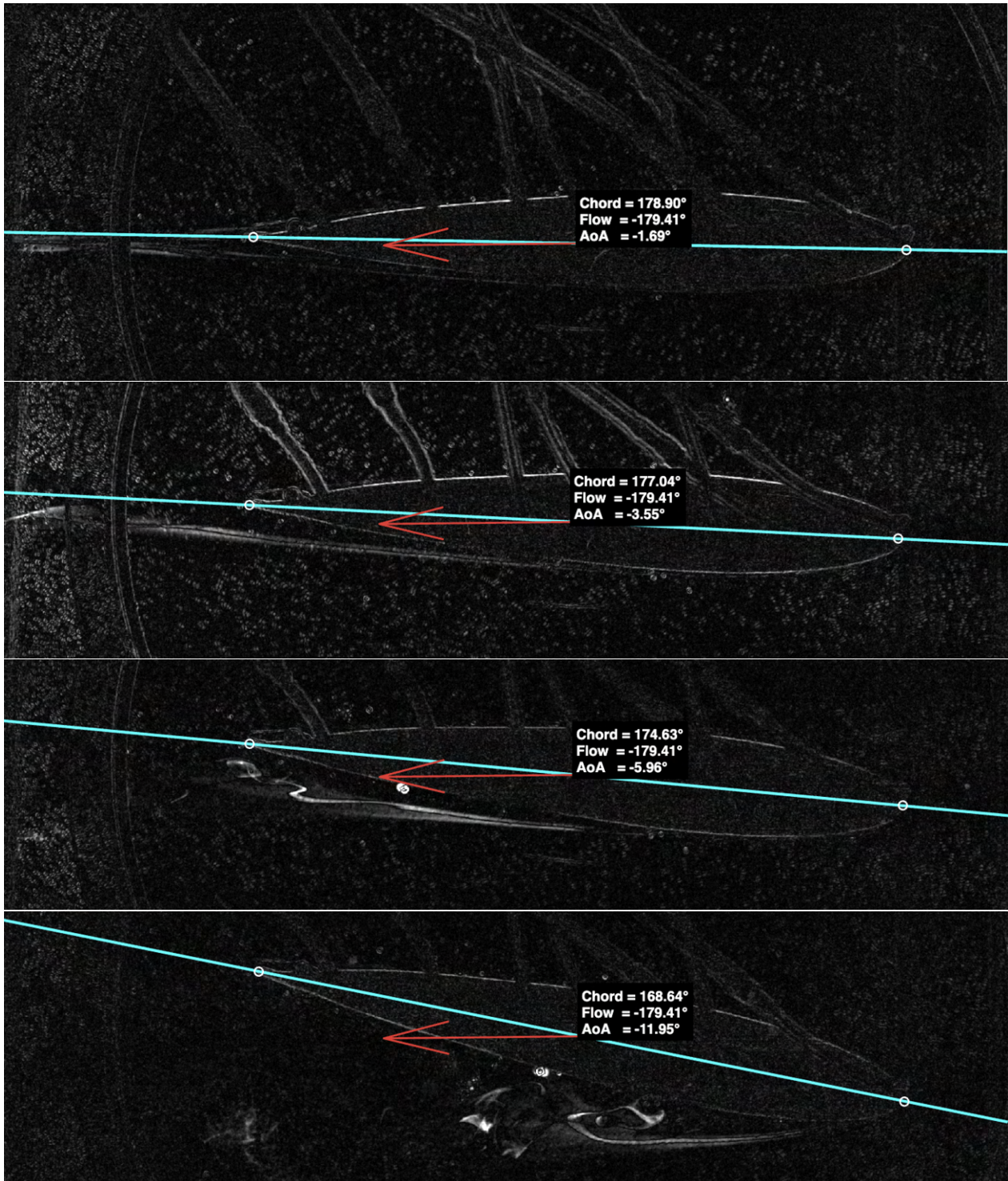


Figure 25: Study of the hydrofoil flow at various angles of attack until stall occurs